

Brazil becomes Associate Member State of CERN

Brazil has become the first Associate Member State of CERN in the Americas



Brazil has become the first Associate Member State of CERN in the Americas, following official notification that the country has completed its internal approval procedures in respect of the agreement signed in March 2022 granting it that status and of the Protocol on Privileges and Immunities of the Organization. The starting date of Brazil's status as an Associate Member State is 13 March 2024.

Formal cooperation between CERN and Brazil started in 1990 with the signature of an International Cooperation Agreement, allowing Brazilian researchers to participate in the DELPHI experiment at the Large Electron–Positron Collider (LEP). Over the past decade, Brazil's experimental particle-physics community has doubled in size. At the four main Large Hadron Collider (LHC) experiments alone, about 200 Brazilian scientists, engineers and students collaborate in fields ranging from hardware and data processing to physics analysis.

Today, Brazilian institutes participate in all the main experiments at the LHC – ALICE, ATLAS, CMS and LHCb and their ongoing and planned upgrades – as well as in ALPHA at the anti-proton decelerator. They are also involved in experiments at ISOLDE, ProtoDUNE at the Neutrino Platform and instrumentation projects such as Medipix.

>>> *Continued at the bottom of page 2*

A Word from Fabiola Gianotti

News from the March CERN Council Session.....p.2

Contents

News.....	p.1-14
Brazil becomes Associate Member State of CERN	
Accelerator Report: Protons or Easter eggs? Let's hope for both	
CERN70: Tracing Particles	
News from Moriond: First observation of photons-to-taus in proton–proton collisions by CMS	
News from Moriond: The delicate balance of lepton flavours	
Spring at CERN, your photos	
Scientists use n_TOF to investigate how cerium is produced in the Universe	
Observing accelerator resonances in 4D	
CERN publishes knowledge transfer highlights from 2023	
CERN and STFC support environmentally sustainable physics	
CERN launches the White Rabbit Collaboration World Wide Web at 35	
CERN rewarded for its contributions to cloud computing	
Computer Security: Day of the open firewall	

Official News.....	p.14-17
Staff Rules and Regulations - 11th Edition	
Administrative Circular No. 33 (Rev.) – Duty Travel	
Arrival/change of address in Switzerland and Swiss “Attestation de départ”	
Composition of the Joint Advisory Disciplinary Board (JADB / CPCD)	
Composition of the Joint Advisory Appeals Board (JAAB / CPCR)	

Announcements.....	p.17-19
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Ombud's Corner.....	p.20
Difficult conversations (part I): “what happened?”	

A Word from Fabiola Gianotti

News from the March CERN Council Session

The successful restart of the accelerator complex, Brazil's Associate Membership and the next update of the European Strategy for Particle Physics were among many topics discussed in the 216th session of the CERN Council

On 22 March, the CERN Council concluded a very fruitful March Session. The delegates congratulated CERN and its personnel on the successful year-end technical stop and restart of the accelerator complex and the experiments, and on the excellent progress made across the full spectrum of Laboratory activities, as was also demonstrated by the depth and breadth of the 2023 Annual Progress Report.

During presentations of the scientific programme, the Council delegates were informed that the SHiP experiment, which will search for feebly interacting particles, has been chosen to run in the North Area's ECN3 hall using high-intensity proton beams from the SPS. This decision concluded a process that took more than a year, involving the Physics Beyond Colliders study group and the SPS and PS Experiments Committee (SPSC).

The Council warmly welcomed [Brazil](#), the first country in Latin America to join CERN, as an Associate Member State.

Looking towards the future, the Council approved the overall timeline for the next update of the [European Strategy for Particle Physics](#). The deadline for the submission of community input will be 31 March 2025, and an email on the subject will be sent soon by the President of Council to the community. The goal is to conclude the process in June 2026, with the Strategy updated by the Council. A more detailed timeline will be established over the coming months.

The lively and very productive March Council Session again demonstrated the strong and sustained support of our Member and Associate Member States for CERN's mission and community. We are very grateful to them. On behalf of the Directorate, I also express our heartfelt thanks to the members of the personnel for your hard work and dedication.

Fabiola Gianotti
CERN Director-General

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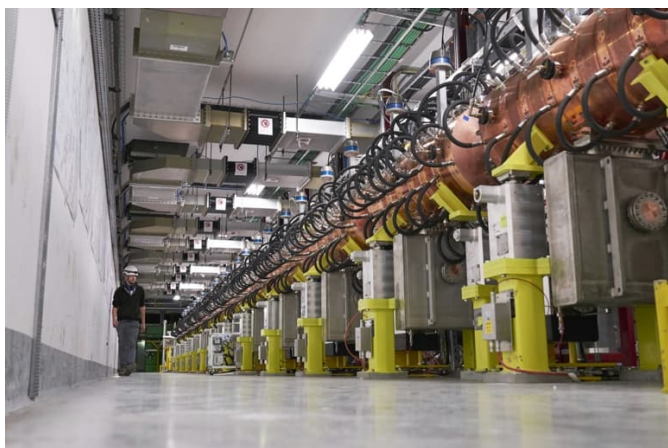
Following on from their participation in the RD51 collaboration, Brazilian teams are also contributing to setting up the DRD1 and DRD3 R&D collaborations for future detectors. Brazilian nationals also participate very actively in CERN training and outreach programmes.

Beyond particle physics, CERN and Brazil's National Centre for Research in Energy and Materials (CNPEM) have also been formally

cooperating since December 2020 on accelerator technology R&D and its applications.

As an Associate Member State, Brazil is entitled to appoint representatives to attend meetings of the CERN Council and the Finance Committee. Its nationals are eligible to apply for limited-duration staff positions and CERN's graduate programmes, and its industry is entitled to bid for CERN contracts, increasing opportunities for industrial collaboration in advanced technologies.

Accelerator Report: Protons or Easter eggs? Let's hope for both



The 86-metre-long Linac4 is composed of different accelerating structures, including the Cell-Coupled-Drift-Tube-Linacs (CCDTLs). (Image: CERN)

Beam commissioning is progressing well across the entire accelerator complex, with initial completion achieved in the first machines of the chain. Last week, the first physics experiments started in the East Area, behind the PS, and others will follow suit shortly.

However, despite the overall positive momentum of beam commissioning, challenges have arisen along the way, highlighting the complexities involved. Last week, some of the components of one of the Linac4 accelerating structures, specifically the Cell-Coupled-Drift-Tube-Linacs (CCDTLs) 3 and 4, presented some issues. Both CCDTLs rely on a single klystron*, a high-power microwave amplifier crucial for providing accelerating power to the structures, which, in turn, transfer the power to the protons, which are then accelerated.

The high-voltage and high-frequency amplifier chain, including the klystron, experienced frequent voltage breakdowns, resulting in a significant drop in accelerating voltage within the two CCDTLs. This disruption severely perturbed the beam, rendering it unusable for the PS Booster. Experts intervened multiple times, initially focusing on fine-tuning the parameters of the amplifier chain and later on cleaning and replacing various components suspected to be causing the breakdowns.

By 22 March, a set of parameters was established to allow the acceleration of beams with a low proton intensity, enabling commissioning activities to continue in the downstream

machines, including the LHC. However, these parameters did not meet the requirements for generating the full-blown physics beams that will be required in the coming weeks. On 24 March, a collaborative effort with experts from various groups convened in the CERN Control Centre (CCC) to conduct a final assessment. This evaluation aimed to determine whether the klystron needed replacing.

After re-establishing the parameters suitable for high-intensity beam acceleration, the beam was switched back on. Unfortunately, within the first hour, at least two high-voltage breakdowns occurred – the team thus concluded that the klystron replacement was necessary.

To maintain commissioning activities in the downstream machines, parameters allowing low-intensity beam acceleration were reinstated. This allowed operations to continue until Monday morning, when the klystron replacement process started. Such an intervention typically requires two to three days before beam operations can be restored.

Meanwhile, commissioning activities in the downstream machines have been suspended and the start of physics at the n_TOF facility, behind the PS, has been postponed (it was originally scheduled to start on 25 March). Commissioning of the North Area's secondary beams began on 22 March instead of 25 March. Thanks to this head start, the incident in Linac4 does not impact the overall schedule for the North Area, where physics is still scheduled to start on 10 April.

In Linac4, the old klystron has been removed, and the new one had been installed and tested by 26 March. Beam was sent to the PS Booster at 5 p.m. that day and, at 9 p.m., the LHC beam commissioning activities resumed. Since then, they have been progressing well.

On 27 March, beams entered into “test” collisions at the target energy of 6.8 TeV in the LHC. These were not yet stable beams, which meant that the experiments did not take data. Collisions for physics at 6.8 TeV are expected to take place on 8 April.

Rende Steerenberg

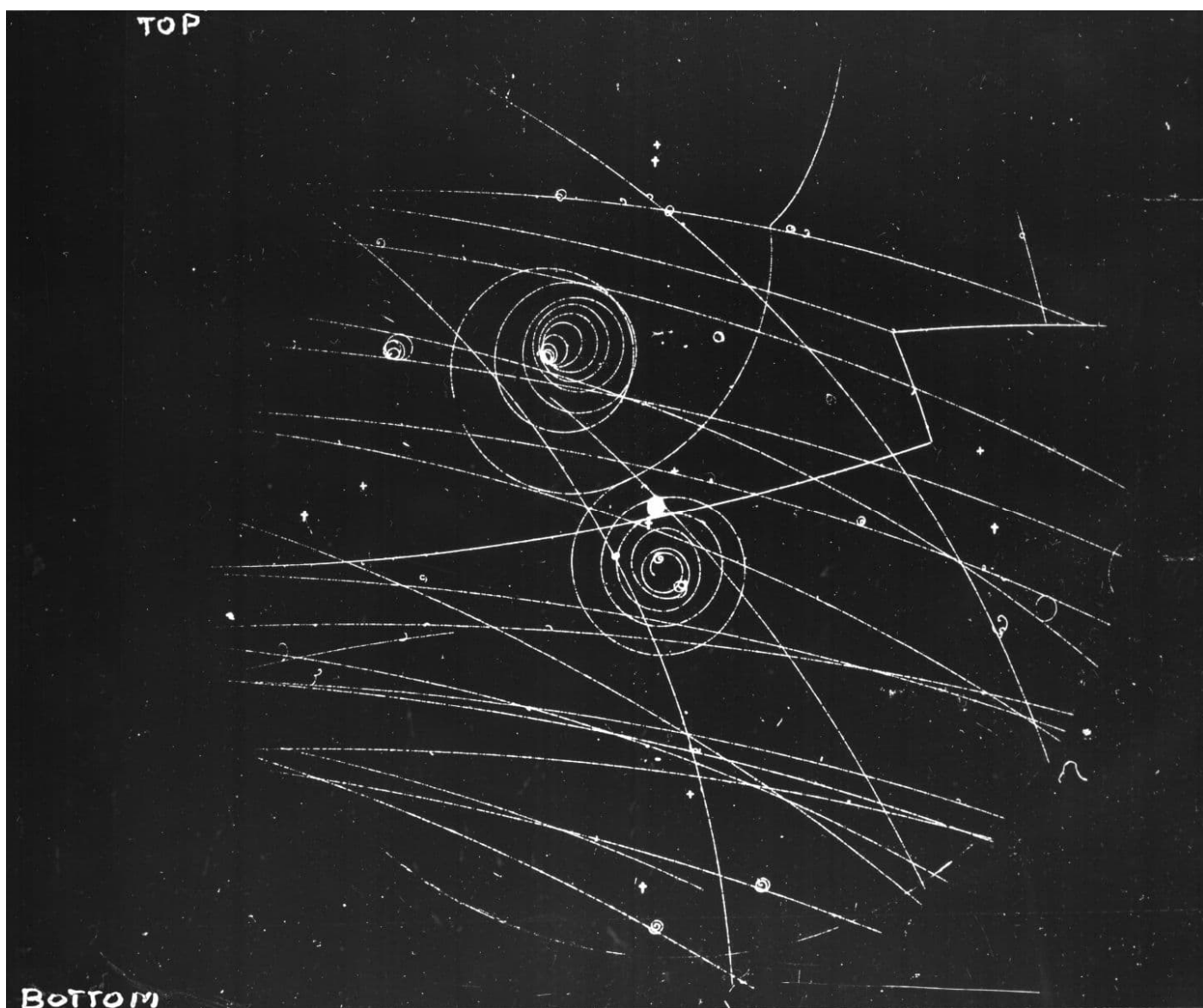
** A klystron is a high-power microwave amplifier used to generate high-power radiofrequency (RF) signals at a specific frequency. It operates on the principle of velocity modulation, where bunches of*

electrons are alternately accelerated and decelerated within a resonant cavity structure. This modulation process results in the amplification of the RF signal.

CERN70: Tracing particles

Part 6 of the CERN70 series. Madeleine Znoy was one of the people responsible for “scanning” the films from the bubble chambers for interesting events

In the 1960s and 1970s, two techniques for accurately recording the tracks of invisible particles dominated experimental high-energy physics, the bubble chamber and the spark chamber. The pictures produced – simple photographs – were then examined for interesting tracks by specially trained personnel, the “scanners”.



Bubble chamber photo from 1961 showing particle tracks. (Image: CERN)

The bubble chamber programme at CERN started in 1959, when the 30-centimetre hydrogen bubble chamber recorded its first image. In parallel, a new, more ambitious project had begun: the construction of

a two-metre chamber, which started operating in 1965. In 12 years of active life, this “big” bubble chamber produced more than 40 million photographs and used 20 000 km of film (enough to go half-way around the Earth!).

CERN also pioneered the use of bubble chambers for the study of neutrino interactions, and it was the great interest in this field that led to the construction of the Big European Bubble Chamber (BEBC), with 20 cubic metres of liquid hydrogen, and Gargamelle, which contributed to the outstanding success story of the discovery of weak neutral currents at CERN.

The acquired expertise in engineering and cryogenics served the construction of large detectors for experiments at the Large Electron-Positron Collider (LEP) and the Large Hadron Collider (LHC), and the collaborations established then were the starting point for the international cooperations that prevails today.

Recollections

Madeleine Znoy was a “scanner” at CERN, a person responsible to examine bubble chamber films to identify interesting events. She went on to work for the OPAL experiment at LEP, and then for the electronics coordination team of the ATLAS experiment at the LHC.

“Our work as “scanners” consisted of identifying the trajectories of the particles in the images taken inside the bubble chambers. The scanning was carried out on the basis of criteria laid down by the physicists. Initially, the work was done manually, using a pencil and a sheet of paper to note down the coordinates of where the interactions had taken place as one would on a map (e.g. A-3, B-4) and a description of the interactions (the number of tracks, disintegration, ionisation, energy, etc.).



Madeleine Znoy at a scanning table in 1976 with a projection of a Big European Bubble Chamber (BEBC) picture. (Image: CERN)

Later, the measuring equipment evolved and was linked up to a computer. The data were recorded and immediately transmitted to be processed by a reconstruction programme which would send

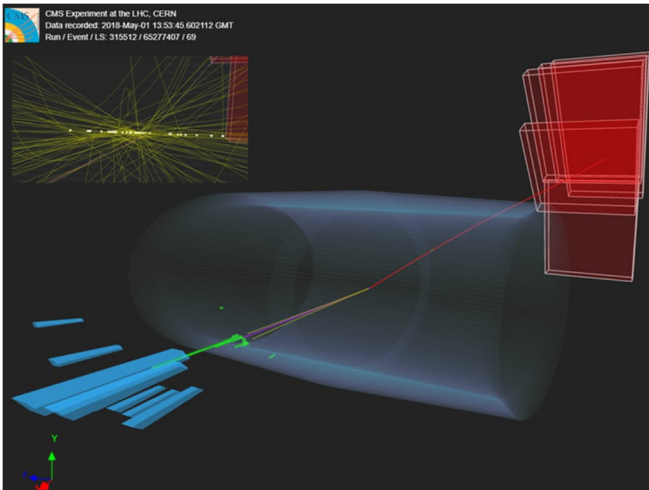
back a message requesting a further set of measurements, corrections to be made or indicating that it had all the information it required.

“Scanning” took place round the clock, as the quantities of film to get through were enormous. Female scanners examined the images between 7 a.m. and 10 p.m. and men, often students, would take over between 10 p.m. and 7 a.m. The shifts only lasted four hours because the work put a lot of strain on the eyes. We worked in complete darkness, with three projectors to illuminate the films we had to examine. Each event was actually analysed by three cameras, from three different angles. We could scan hundreds of images in each shift. I even beat a record by evaluating 750 images in four hours! At first, the physicists thought it was impossible and that I must have missed some interesting events. In reality, everything was correct and they were very surprised!

It was true detective work. There was always something to learn. “Scanning” the trajectories of the elementary particles recorded in the images allowed us the opportunity to gain an insight into that type of physics research at our work station. By virtue of always participating in the initial phases of the experiments, I would take part in the development and testing of new measurement systems. Once a system went into production, I would move on to the next experiment, working on another system with another team. I really enjoyed working with physicists and technicians as well as taking part in the collaboration with other laboratories. We were young and full of enthusiasm. We worked hard but we also knew how to have a good time. Those were good times ...”

News from Moriond: First observation of photons-to-taus in proton–proton collisions by CMS

Using the outstanding tracking capabilities of the CMS detector, the collaboration observed two photons producing two tau-leptons in proton–proton collisions



Recreated candidate event of a $\gamma\gamma \rightarrow \tau\tau$ process in proton–proton collisions measured by the CMS detector. The tau can decay into a muon (red), charged pions (yellow) and neutrinos (not visible); energy deposits in the electromagnetic calorimeter in green and in the hadronic calorimeter in cyan. (Credit: CMS collaboration.)

In March 2024, the CMS collaboration announced the observation of two photons creating two tau leptons in proton–proton collisions. It is the first time that this process has been seen in proton–proton collisions, which was made possible by using the precise tracking capabilities of the CMS detector. It is also the most precise measurement of the tau’s anomalous magnetic moment and offers a new way to constrain the existence of new physics.

The tau, sometimes called tauon, is a peculiar particle in the family of leptons. In general, leptons, together with quarks, make up the “matter” content of the Standard Model (SM). The tau was only discovered in the late 1970s at SLAC, and its associated neutrino – the tau neutrino – completed the tangible matter part upon its discovery in 2000 by the DONUT collaboration at Fermilab. Precise research for the tau is rather tricky though, as its lifetime is very short: it

remains stable for only $290 \cdot 10^{-15}$ s (a hundred quadrillionth of a second).

The two other charged leptons, the electron and the muon, are rather well studied. A lot is also known about their magnetic moments and their associated anomalous magnetic moments. The former can be understood as the strength and orientation of an imaginary bar magnet inside a particle. This measurable quantity, however, needs corrections at the quantum level arising from virtual particles tugging at the magnetic moment, deviating it from the predicted value. The quantum correction, referred to as anomalous magnetic moment, is of the order of 0.1%. If the theoretical and experimental results disagree, then this anomalous magnetic moment, a_τ , opens doors to physics beyond the SM.

The anomalous magnetic moment of the electron is one of the most precisely known quantities in particle physics and agrees perfectly with the SM. Its muonic counterpart, on the other hand, is one of the most investigated ones, into which research is ongoing. Although theory and experiments have mostly agreed so far, recent results give rise to a tension that requires further investigation.

For the tau, however, the race is still going. It is especially hard to measure its anomalous magnetic moment, a_τ , due to the tau’s short lifetime. The first attempts to measure a_τ after the tau’s discovery came with an uncertainty that was 30 times higher than the size of the quantum corrections. Experimental efforts at CERN with the LEP and LHC detectors improved the constraints, reducing the uncertainties to 20 times the size of the quantum corrections.

In collisions, researchers look for a special process: two photons interacting to produce two tau leptons, also called a di-tau pair, which then decay

into muons, electrons, or charged pions, and neutrinos. So far both ATLAS and CMS have observed this in ultra-peripheral lead–lead collisions. Now, CMS reports on the first observation of the same process during proton–proton collisions. These collisions offer a higher sensitivity to physics beyond the SM as new physics effects increase with the collision energy. With the outstanding tracking capabilities of the CMS detector, the collaboration was able to isolate this specific process from others, by selecting events where the taus are produced without any other track within distances as small as 1 mm. “This remarkable achievement of detecting ultra-peripheral proton–proton collisions sets the stage for many groundbreaking measurements of this kind with the CMS experiment,” said Michael Pitt, from the CMS analysis team.

This new method offers a new way to constrain the tau anomalous magnetic moment, which the CMS collaboration tried out immediately. While the significance will be improved with future run data, their new measurement places the tightest constraints so far, with higher precision than ever before. It reduces the uncertainty from the predictions down to only three times the size of the quantum corrections. “It is truly exciting that we can finally narrow down some of the basic properties of the elusive tau lepton,” said Izaak Neutelings, from the CMS analysis team. “This analysis introduces a novel approach to probe tau $g-2$ and revitalises measurements that have remained stagnant for more than two decades,” added Xuelong Qin, another member of the analysis team.

More to come from Moriond in [physics news](#)

News from Moriond: The delicate balance of lepton flavours

New ATLAS measurement of lepton flavour universality, a key feature of the Standard Model, in decays of the W boson is more precise than all previous measurements combined

In a talk at the ongoing Rencontres de Moriond conference, the ATLAS collaboration presented the result of its latest test of a key principle of the Standard Model of particle physics known as lepton flavour universality. The precision of the result is the best yet achieved by a single experiment in decays of the W boson and surpasses that of the current experimental average.

Most elementary particles can be classed into groups or families with similar properties. For example, the lepton family includes the electron, which forms the negatively charged cloud of particles surrounding the nucleus in every atom, the muon, a heavier particle found in cosmic rays, and the tau-lepton, an even heavier short-lived particle only seen in high-energy particle interactions.

As far as physicists know, the only difference between these particles is their mass, as generated through their different strengths of interaction with the fundamental field associated with the Higgs boson. In particular, a remarkable feature of the Standard Model is that each lepton

type, or “flavour”, is equally likely to interact with a W boson, the electrically charged carrier of the weak force that is one of the four fundamental forces of nature. This principle is known as lepton flavour universality.

High-precision tests of lepton flavour universality, as obtained by comparing the rates of decay of the W boson into an electron and an electron neutrino, into a muon and a muon neutrino or into a tau-lepton and a tau neutrino, are therefore sensitive probes of physics beyond the Standard Model. Indeed, if lepton flavour universality holds, these decay rates should be equal (within negligible mass-dependent corrections).

This can be tested by measuring the ratios of the W boson’s rates of decay into the different lepton flavours. One of the challenges associated with such measurements at the Large Hadron Collider (LHC) is the collection of a pure (“unbiased”) sample of W bosons. In a paper released by Nature Physics in 2021, ATLAS reported the world’s most precise measurement of the ratio of the W boson’s rate of decay into a tau-lepton versus its rate of decay into a muon, demonstrating that collision

events in which a pair of top quarks is produced provide an abundant and clean sample of W bosons.

In a recent paper, ATLAS released a new measurement, this time addressing the ratio of the W boson's rate of decay into a muon versus its rate of decay into an electron. While the combination of all previous measurements showed that this ratio is within about 0.6% of unity, corresponding to equal decay rates, there was still room for improvement.

The new ATLAS result is based on a study of its full dataset from the second run of the LHC, collected between 2015 and 2018. The analysis looked at over 100 million top-quark-pair collision events. The top quark decays promptly into a W boson and a bottom quark, so this sample provides 100 million pairs of W bosons. By counting the number of these events with two electrons (and no muon) or two muons (and no electron), physicists can test whether the W boson decays more often into an electron or a muon.

However, it's not that simple. The Z boson, the electrically neutral carrier of the weak force, can also decay into a pair of electrons or muons, leaving a similar experimental signature to that of a top-quark pair. Since the combined mass of the leptons in Z-boson events clusters around the Z-

boson mass of 91 GeV, this background process can be estimated and subtracted.

Moreover, as a result of measurements conducted in the 1990s at CERN's Large Electron-Positron (LEP) collider, the LHC's predecessor, and at the Stanford Linear Collider (SLC), the ratio of the Z boson's rate of decay into two muons versus its rate of decay into two electrons is known to be equal to unity within 0.3%. Thus, in this ATLAS analysis, the Z boson's decay rate ratio was determined as a reference measurement, allowing researchers to reduce uncertainties coming from the reconstruction of electrons and muons. Additionally, as many measurement uncertainties are similar in the events with two electrons and those with two muons, they were found to have only a minor effect on the measured decay rate ratio.

The final result from this new ATLAS analysis is a ratio of 0.9995, with an uncertainty of 0.0045, perfectly compatible with unity. With an uncertainty of only 0.45%, the result is more precise than all previous measurements combined (see figure below). For now, lepton flavour universality survives intact.

ATLAS collaboration

More to come from Moriond in [physics news](#)

Spring at CERN, your photos



Ice-white blossom in front of ISOLDE (Image: Sanje Fenkart/CERN)

Blossom, blue sky and buildings proved the winning combination for our "spring at CERN" photo competition. Congratulations to Sanje Fenkart from the IR department, who wins not just one CAGI Chocopass, but two, allowing her and a friend to spend a day exploring Geneva and tasting from a range of chocolate shops.

Thank you to all of you who sent in your photos. They are beautiful and are now available in a [CC-BY photo collection](#).

We'd like to thank the International Geneva Welcome Centre (CAGI) for their sweet (pun intended) gesture of offering these prizes. The CAGI cultural kiosk is located in CERN's main building and is open from Monday to Friday from 8:30 a.m. to 11:00 a.m. and from 11:30 a.m. to 2:30 p.m. It offers numerous discounts for local

Scientists use n_TOF to investigate how cerium is produced in the Universe

The CERN facility allowed scientists to see how this rare Earth metal is synthesised in stars. The results open up new questions about stellar nucleosynthesis and the chemical evolution of galaxies

Cerium is a rare Earth metal that has numerous technological applications, for example in some types of lightbulbs and flat-screen TVs. While the element is rare in Earth's crust, it is slightly more abundant in the Universe. However, much is unknown about how it is synthesised in stars. Now, in a new study published in [Physical Review Letters](#), the n_TOF collaboration at CERN investigates how cerium is produced in stars. The results differ from what was expected from theory, indicating a need to review the mechanisms believed to be responsible for the production of cerium – and other heavier elements – in the Universe.

“The measurement we carried out enabled us to identify nuclear resonances never observed before in the energy range involved in the production of cerium in stars,” explains Simone Amaducci, of INFN's Southern National Laboratories and first author of the study. “This is thanks to the very-high-energy resolution of the experimental apparatus at CERN and the availability of a very pure sample of cerium 140.” The abundance of elements heavier than iron observed in stars (such as tin, silver, gold and lead) can be reproduced mathematically by hypothesising the existence of two neutron capture processes: the slow (s) process and the rapid (r) process. The s process corresponds to a neutron flux of 10 million neutrons per cubic centimetre while the r process has a flux of more than one million billion neutrons per cubic centimetre. The s process is theorised to produce about half of the elements heavier than iron in the Universe, including cerium.

CERN's Neutron Time-of-Flight facility (n_TOF) is designed to study neutron interactions, such as

those that occur in stars. In this study, the scientists used the facility to measure the nuclear reaction of the cerium 140 isotope with a neutron to produce isotope 141. According to sophisticated theoretical models, this particular reaction plays a crucial role in the synthesis of heavy elements in stars. Specifically, the scientists looked at the reaction's cross section: the physical quantity that expresses the probability that a reaction occurs. The scientists measured the cross section at a wide range of energies with an accuracy 5% higher than previous measurements. The results open up new questions about the chemical composition of the Universe. “What intrigued us at the beginning was a discrepancy between theoretical star models and observational data of cerium in the stars of the M22 globular cluster in the Sagittarius constellation,” explains Sergio Cristallo of INAF's Abruzzo Astronomical Observatory, who proposed the experiment. “The new nuclear data differs significantly, up to 40%, from the data present in the nuclear databases currently used, definitely beyond the estimated uncertainty.”

These results have notable astrophysical implications, suggesting a 20% reduction in the contribution of the s process to the abundance of cerium in the Universe. This means a paradigm shift is required in the theory of cerium nucleosynthesis: other physical processes that are not currently included would need to be considered in calculations of stellar evolution. Furthermore, the new data has a significant impact on scientists' understanding of the chemical evolution of galaxies, which also affects the production of heavier elements in the Universe.

Observing accelerator resonances in 4D

For the first time, scientists at CERN, in collaboration with scientists at GSI, have been able to measure a coupled resonance structure that may cause particle loss in accelerators

Whether in listening to music or pushing a swing in the playground, we are all familiar with resonances and how they amplify an effect – a sound or a movement, for example. However, in high-intensity circular particle accelerators, resonances can be an inconvenience, causing particles to fly off their course and resulting in beam loss. Predicting how resonances and non-linear phenomena affect particle beams requires some very complex dynamics to be disentangled. For the first time, scientists at the Super Proton Synchrotron (SPS), in collaboration with scientists at GSI in Darmstadt, have been able to experimentally prove the existence of a particular resonance structure. While it had previously been theorised and appeared in simulations, this structure is very difficult to study experimentally as it affects particles in a four dimensional space*. These latest results, published in Nature Physics, will help to improve the beam quality for low-energy and high-brightness beams for the LHC injectors at CERN and the SIS18/SIS100 facility at GSI, as well as for high-energy beams with large luminosity, such as the LHC and future high-energy colliders.

“With these resonances, what happens is that particles don’t follow exactly the path we want and then fly away and get lost,” says Giuliano Franchetti, a scientist at GSI and one of the paper’s authors. “This causes beam degradation and makes it difficult to reach the required beam parameters.”

The idea to look for the cause of this emerged in 2002, when scientists at GSI and CERN realised that particle losses increased as accelerators pushed for higher beam intensity. “The collaboration came from the need to understand what was limiting these machines so that we could deliver the beam performance and intensity needed for the future,” says Hannes Bartosik, a scientist at CERN and another of the paper’s authors.

Over many years, theories and simulations were developed to understand how resonances affected particle motion in high-intensity beams. “It required an enormous simulation effort by

large accelerator teams to understand the effect of the resonances on beam stability,” says Frank Schmidt at CERN, also one of the paper’s authors. The simulations showed that resonance structures induced by coupling in two degrees of freedom are one of the main causes of beam degradation.

It took a long time to devise how to look for these resonance structures experimentally. This is because they are four dimensional* and require the beam to be measured in both the horizontal and the vertical planes to see if they exist. “In accelerator physics, the thinking is often in only one plane,” adds Franchetti.

To measure how resonances affect particle motion, the scientists used beam position monitors around the SPS. Over approximately 3000 beam passages, the monitors measured whether the particles in the beam were centred or more to one side, in both the horizontal and vertical planes. The resonance structure that was found is shown in Figure 1.

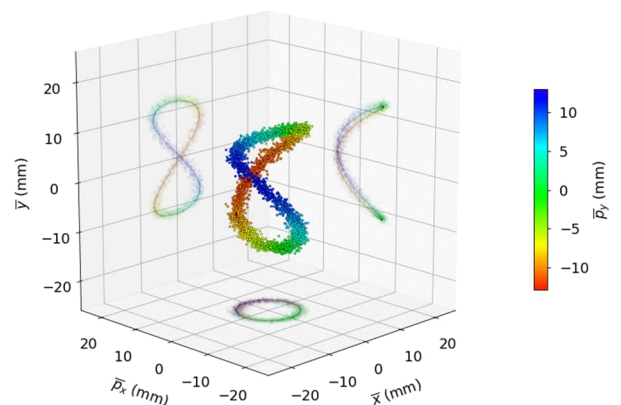


Figure 1: Conceptually visualising 4D resonance structures is much more complicated than one-dimensional resonances. This image shows the 4D resonance structure measured in the SPS. (Image: H. Bartosik, G. Franchetti and F. Schmidt, Nature Physics)

“What makes our recent finding so special is that it shows how individual particles behave in a coupled resonance,” continues Bartosik. “We can demonstrate that the experimental findings agree with what had been predicted based on theory and simulation.”

While the existence of the coupled resonance structures has now been observed experimentally, much more remains to be done to reduce their detrimental effect. “We’re developing a theory to describe how particles move in the presence of these resonances,” continues Franchetti. “With this study, coupled with all the previous ones, we hope we will get clues on how to avoid or minimise

the effects of these resonances for current and future accelerators.”

Naomi Dinmore

**Space refers to “phase space” – the space in which all possible states of a system are represented*

CERN publishes knowledge transfer highlights from 2023

[CERN's new digital report “Accelerating Innovation Through Partnerships”](#) highlights knowledge transfer activities from 2023. It showcases concrete applications of CERN technologies and know-how, with diverse examples in the [healthcare](#), [environment](#), [aerospace](#), [digital](#) and [quantum](#) fields.

Find out more about CERN's ongoing partnerships with industry, academia, research institutions and hospitals in its Member and Associate Member States. See how entrepreneurs are supported by

the recently launched CERN Venture Connect programme. Discover how these activities not only drive innovation but also have a positive impact on society.

Register now for the public event “[The virtuous circle of knowledge and innovation](#)” taking place on 18 April 2024 in CERN Science Gateway as part of the series of [events for CERN's 70th birthday](#). For more information about CERN's ongoing knowledge transfer activities, visit [kt.cern](#).

CERN Knowledge Transfer Group

CERN and STFC support environmentally sustainable physics

CERN and the UK's Science and Technology Facilities Council (STFC) sign a new agreement to support the development of more sustainable particle accelerators

On 22 March, CERN and the UK's Science and Technology Facilities Council (STFC) signed a new agreement to collaborate on the research and development of advanced new technologies to make future particle accelerators significantly more sustainable.

Minimising the environmental impact of particle physics activities and ensuring their sustainability and energy efficiency is one of the key recommendations in the last update of the [European Strategy for Particle Physics](#), published in 2020.

“CERN is [fully committed](#) to fostering sustainability across its existing and forthcoming projects, actively engaging in a variety of [initiatives](#),” explains Mike Lamont, CERN Director for Accelerators and Technology. “These include sourcing renewable energy, implementing heat recovery schemes and forging collaborations

with industry to explore innovative applications of sustainable technology, such as high-power electricity distribution in various contexts. Our philosophy in this regard aligns well with that of the STFC and we look forward to exploiting the potential of this collaboration – together we are stronger.”

The agreement will act as a framework to better direct CERN and STFC's funding, expertise and technological investment to minimise environmental impact. It provides guidance and recommendations that consider the entire lifecycle of accelerator facilities, from design and construction to operation and decommissioning. The agreement also outlines a proposal for STFC to establish a new Centre of Excellence in Sustainable Accelerators (CESA) at the Daresbury Laboratory in the UK. CESA would conduct original research in sustainable accelerator technologies and train

accelerator scientists, technicians and engineers in the skills required to develop new accelerators with sustainability at the heart of the design.

For more details, see the [UKRI website](#).

CERN launches the White Rabbit Collaboration

White Rabbit, an open-source timing technology developed at CERN, has applications far beyond particle physics. To foster its uptake by industry, CERN has launched the White Rabbit Collaboration

[White Rabbit](#) (WR) is a technology developed at CERN, in collaboration with institutes and companies, to synchronise devices in the accelerators down to sub-nanoseconds and solve the challenge of establishing a common notion of time across a network. Indeed, at a scale of billionths of a second, the time light takes to travel through a fibre-optic cable and the time the electronics take to process the signal are no longer negligible. To avoid potential delays, the co-inventors of White Rabbit designed a new ethernet switch.

First used in 2012, the application of this fully open-source technology has quickly expanded outside the field of particle physics. In 2020, [it was included](#) in the worldwide industry standard known as [Precision Time Protocol](#) (PTP), governed by the Institute of Electrical and Electronics Engineers ([IEEE](#)).

What's more, CERN recently launched the White Rabbit Collaboration, a membership-based global community whose objective is to maintain a high-performance open-source technology that meets the needs of users and to facilitate its uptake by industry. The WR Collaboration will provide dedicated support and training, facilitate R&D projects between entities with common interests and complementary expertise and establish a testing ecosystem fostering trust in products that incorporate the open-source technology. At CERN, the WR Collaboration Bureau – a dedicated team composed of senior White Rabbit engineers and a community coordinator – will facilitate the day-to-day running of the Collaboration's activities and support its members.

"A key distinctive feature of White Rabbit, as opposed to other technologies that have been developed since, is that it is open source and based on standards", says Javier Serrano, Chair of the White Rabbit Collaboration Board and co-

inventor of the technology. Companies and institutes can therefore adapt it to their needs and incorporate it in their products and systems, while the technology, in turn, benefits from a large community of developers. "The first step to foster industry uptake was to include WR concepts in the IEEE standard. Through this endeavour, we established many links with industry," says Maciej Lipinski, Chair of the White Rabbit Collaboration Council and senior White Rabbit engineer at CERN. White Rabbit is used in the finance sector as well as in many research infrastructures, and it is currently being evaluated for application in the future quantum internet. The technology could also play a key role in the future landscape of global time dissemination technologies, which currently rely heavily on satellites. The infrastructure for the dissemination of time is essential to the economy and underpins most critical national infrastructure. Governments and industry across the globe are therefore [striving to find alternatives to distribute a reference time](#), such as the one WR could offer via optical fibre, with telecom and power grid companies starting to test WR in their networks.

The WR Collaboration comes at a moment when many sectors are undergoing a profound transformation with regards to their timing technology. "The WR Collaboration will provide a neutral gathering point around this open-source technology and define a long-term common vision, establishing a solid ground from which innovation can thrive," continues Amanda Diez Fernandez from CERN's [Knowledge Transfer](#) group and the White Rabbit Community Coordinator.

To learn more about how to join the White Rabbit Collaboration, go to: www.white-rabbit.tech

From 21–22 March, the White Rabbit community met at CERN to celebrate the launch of the WR

Collaboration. To find out more, listen to the recorded talks of the event.

World Wide Web at 35

Thirty-five years ago, a young computer expert working at CERN wrote a proposal that combined accessing information with a desire for broad connectivity and openness. This proposal went on to become the [World Wide Web](#) (WWW), whose impact on society has been profound.

Sir Tim Berners-Lee's [first proposal](#) in March 1989 was for an internet-based hypertext system to link and access information across different computers. In November 1990, this “web of information nodes in which the user can browse at will” was formalised as a proposal, “[WorldWideWeb: Proposal for a HyperText Project](#)”, by Berners-Lee, together with a CERN colleague, Robert Cailliau. By Christmas that year, Berners-Lee had implemented key components,

namely html, http and URL, and created the first Web server, browser and editor (WorldWideWeb). This server is now exhibited in the Laboratory's new visitor centre, [CERN Science Gateway](#).

CERN released the WWW software into the public domain on 30 April 1993, making it freely available for anyone to use and improve. This decision encouraged the use of the Web, and society to benefit from it.

Now, thirty-five years since his original proposal, Sir Tim Berners-Lee reflects on the web's trajectory in an [open letter](#) and states how we, as engaged citizens, can “re-shape a digital future that prioritises human well-being, equity, and autonomy”.

CERN rewarded for its contributions to cloud computing

On 21 March, CERN received the “Top End User” special award from the Cloud Native Computing Foundation (CNCF) for its “*forward-looking approach to leveraging cloud native technologies to address future scientific and operational challenges*”. Prior recipients of the “Top End User” CNCF award have included Mercedes-Benz, Spotify and Apple.

Cloud native technologies are software solutions that exploit the basic features of cloud computing – e.g., scalability, flexibility, data resiliency – and allow system engineers to manage complex computing environments effectively and with minimal effort. CERN has been a member of the CNCF community since 2020 and is considered an “end user” because it develops cloud native

technologies internally but does not sell cloud services externally.

CERN contributes to community-driven cloud computing ecosystems, such as the CNCF, but also runs its own private cloud, which is configured and integrated into the CERN network and authentication services. The Organization's cloud ecosystem is heterogeneous and complex, as Taylor Dolezal, head of ecosystem at CNCF confirmed: “CERN's innovative use of cloud native technologies is a shining example of how open source and collaboration can drive cutting-edge research.”

For more details see the [CNCF announcement](#) and the [IT README](#).

Computer Security: Day of the open firewall

With ongoing vulnerability scans of CERN's internet presence performed by an external specialised company, the Computer Security team's plans to perform penetration testing against selected targets visible to the internet, and the possibility of CERN joining a so-called Bug Bounty programme (a *Bulletin* article on this will come soon), we are preparing for an increasingly thorough assessment of the weaknesses, misconfigurations and vulnerabilities inside CERN – on the campus network, the technical network and the networks dedicated to the different experiments.

Given that the CERN networks are many, vast and interconnected in a complex manner, with tens of thousands of registered devices, thousands of them regularly or permanently connected, a large proportion of unmanaged “bring-your-own” devices or unpatchable and inherently vulnerable devices of the Internet of Things, a very large number of heterogeneous virtual machines and containers running arbitrary applications, and about ten thousand websites leading to millions of webpages, vulnerability scanning and penetration testing of such an environment is complex, complicated and tedious. That's why we have decided to lower CERN's outer perimeter firewall protections for 24 hours on the first Monday of next month so that any external third party interested in poking/hacking/breaking into CERN can do so. The open firewall, allowing any incoming traffic, will facilitate not only the work of

the aforementioned external company, but also that of the students affiliated with our [WhiteHat programme](#), Bug Bounty hunters hoping for an entry on our [Kudos page](#) and any other benign or malicious attacker.

As usual, any ethical party probing CERN during those 24 hours is supposed to stop their activity before any damage or destruction is done and to report all their findings immediately to us so that they can be addressed, controlled, mitigated and fixed. For those cases where the scans and tests are performed by malicious actors, our [network-based intrusion detection system](#) connected to the outer perimeter firewall will stay alert and monitor all activities in the hope of identifying their ill-doing well in time, as [we managed to in the past](#). The Computer Security team will, exceptionally, cover its duties 24/7. Of course, we cannot guarantee that no damage will be done by any malicious attacker, but we are counting on the robustness, resilience and up-to-dateness of your systems, devices, virtual machines/containers and websites. This risk is also the reason why we will open the firewall for just 24 hours: this tight time window should keep [any collateral damage](#) low. So, stay tuned for [next Monday, 1 April, 00:00 to 23:59](#), the day when we shall learn more about the security of CERN's internal networks, and subsequently further improve all the systems connected to it.

Computer Security team

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.

Staff Rules and Regulations - 11th Edition

Modification No. 23

In accordance with recommendations made by the Finance Committee and decisions taken by Council in December 2023 (CERN/FC/6747-CERN/3770), please find below the pages of the Staff Rules and Regulations which have been updated further to the modifications coming into force on 1 April 2024:

Chapter V – Financial conditions
Section 1 – Financial benefits (*correction of pages 43bis and 44 and modification of page 45*)

The complete updated electronic version of the Staff Rules and Regulations is accessible via [CDS](#).

HR department

Administrative Circular No. 33 (Rev.) – Duty Travel

Administrative Circular No. 33 (Rev.) entitled "Duty Travel", approved by the Director-General, is now available via the following [link](#).

This revision cancels and replaces Administrative Circular No. 33 entitled "Duty Travel" dated January 2020.

The objective of these modifications is to implement the amendment of Staff Regulation R V 1.27, which came into force on 1 January 2024. This amendment provides for the direct payment of accommodation costs incurred during duty travel or their reimbursement on the basis of actual expenses, within the limits of the

established daily travel allowance, rather than continuing to offer a lump-sum payment option.

The aim is to improve cost-effectiveness and to ensure uniform treatment of all members of the personnel. In exceptional cases, the traveler's department Head may authorise reimbursement in excess of the maximum portion of the daily travel allowance for accommodation, provided this is in the interests of the Organization.

The revised Circular entered into force on 1 March 2024.

HR department

Arrival/change of address in Switzerland and Swiss "Attestation de départ"

The Swiss authorities strongly recommend that members of the personnel declare their arrival in Switzerland and any change of address within 14 days, particularly if they have children of school age.

In addition, certain formalities linked to leaving Switzerland (e.g. lease termination, customs declaration, bank account closure or insurance policy cancellation) may require an "*attestation de départ*" (certificate of departure), which will only be issued to people who have previously declared their arrival.

1. Arrival and change of address in Switzerland

a) Procedure for people living in the Canton of Geneva:

The *Office cantonal de la population et des migrations* (OCPM) should be notified either by post or via the online "e-démarches" procedure (see: <https://www.ge.ch/annoncer-mon-arrivee-office-cantonal-population-migrations/annonce-arrivee-geneve-detenteurs-carte-legitimation> and <https://www.ge.ch/annoncer-changement-adresse-geneve>).

b) Procedure for people living in the Canton of Vaud:

Declaration should be made to the *Contrôle de l'habitant* for the commune in which they live. NB: Holders of a "*carte de légitimation*" (Swiss card) issued by the Swiss Federal Department of Foreign Affairs, except for holders of the "S" card, cannot

use the online “eDéménagement” procedure (see: <https://www.vd.ch/themes/population/control-des-habitants/changement-dadresse>).

2. Declaration of departure and the “attestation de départ”

a) Procedure to follow for people living in the Canton of Geneva:

The declaration of departure should be made either by sending the “[annonce de départ](#)” form and supporting documents to the OCPM by post (Case postale 2652, 1211 Genève 2) or online through an “e-démarches” account (see: <https://www.ge.ch/annoncer-mon-depart-ocpm>).

A declaration of departure does not generate an automatic confirmation but an “attestation de départ” can be requested.

The “attestation de départ” can be requested no sooner than 15 working days after the declaration of departure and will be issued no sooner than one month before the departure date (<https://www.ge.ch/obtenir-attestation-ocpm/commander-attestation-depart>).

The declaration of departure form can be used to list all the family members concerned. However, there is no overall “family” certificate; the “attestation de départ” is for an individual, and if necessary a separate document will have to be requested for each family member.

The “attestation de départ” costs 25 CHF.

It is valid for two months.

b) Procedure to follow for people living in the Canton of Vaud:

Members of the personnel should go in person to the *Contrôle de l’habitant* for the commune where they live with their “carte de légitimation” (Swiss card) issued by the Federal Department of Foreign Affairs in order to declare their departure. Depending which commune they live in, a fee may be due.

Remember, members of the personnel must declare any change in their situation to the Organization within 30 days (particularly concerning their [address](#), [family situation](#) or [nationality](#)).

Host State Relations service

Composition of the Joint Advisory Disciplinary Board (JADB / CPCD)

2024 Exercise

	Appointed by the Director-General	Appointed by the Staff Association
<u>Members</u>	John PYM / DG	Nick ZIOGAS / IPT
<u>1st deputies</u>	Gianluigi ARDUINI / BE	Gunnar LINDELL / HSE
<u>2nd deputies</u>	Rosario PRINCIPE / TE	Nicolas SALOMON / PF

John Pym and Nick Ziogas have drawn up the following list of staff members from among whom the Chairperson of the Board may be chosen when required:

- Sophie BARON / EP
- Kandy MITCHELL / PF
- Johan BREMER / TE

- Alberto PACE / IT
- Ilias EFTHYMIPOULOS / BE
- Laurent TAVIAN / ATS
- Gabriel THIEDE / FAP
- Quentin KING / SY
- Giovanna VANDONI / ATS
- Sebastian LOPIENSKI / IT

Composition of the Joint Advisory Appeals Board (JAAB / CPCR)

2024 Exercise

	Appointed by the Director-General	Appointed by the Staff Association
<u>Members</u>	Valeria PEREZ REALE / TE	François DUVAL / EP
<u>1st deputies</u>	Joël CLOSIER / EP	Nicolas SALOMON / PF
<u>2nd deputies</u>	Raymond VENESS / SY	Silvia SCHUH-ERHARD / BE

Valeria Perez Reale and François Duval have drawn up the following list of staff members from whom the Chairperson of the Board may be chosen when required:

- Jan BORBURGH / SY
- François BRIARD / IR
- Isabel FERNANDEZ GONZALEZ / SCE
- Silvia GRAU / EN
- Nils HOIMYR / IT

- Pedro MARTEL / EN
- Jesper NIELSEN / BE
- Chiara PASQUINO / SY
- Laurent ROY / EP
- Jens VIGEN / RCS

Mediators [see Administrative Circular N° 6 (Rev. 1) entitled "Review procedure"] will also be selected from this list of ten staff members.

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

18 Apr 19:30 | Knowledge Sharing | CERN
Science Gateway | CERN70 | EN
[CERN70 public event: The virtuous circle of knowledge and innovation](#) (more [CERN70](#) events)

24 Apr 18:00 | Knowledge Sharing | Le Globe de la science et de l'innovation | Événements publics | FR
[Session de présentation : Progrès de l'étude de faisabilité FCC](#)

Cet événement public est destiné aux résidents des communautés locales (l'inscription est nécessaire, le nombre de places étant limité). Nous invitons la communauté du CERN à suivre l'événement en ligne (sans inscription).

26 Apr 14:30 | Knowledge Sharing | Online | CERN Alumni | EN
[Moving Out of Academia to Machine Learning](#)

Registration now open

15 May | Knowledge Sharing | Lausanne | Famelab | EN, FR or DE
[Famelab Switzerland local heats](#), open to 18–35-year-olds

23 May 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN
[L&D Micro-talk 10 – Artificial Intelligence \(AI\) & The Future of Work](#)

2–15 Jun | Accelerators | Sint-Michielsgestel (NL) | CERN Accelerator School | EN

[Mechanical & Materials Engineering for Particle Accelerators and Detectors](#)

10–12 Jun | Knowledge Sharing | Online | Sustainable HEP | EN
[Sustainable HEP 2024 – 3rd Edition](#)

11–13 Jun | Engineering | CERN | FDF | EN
[1st FPGA \(Field Programmable Gate Arrays\) Developers' Forum meeting](#)

12–25 Jun | Physics | Nakhon Pathom (TH) | Asia-Europe-Pacific School of High-Energy Physics (AEPSHEP) | EN
[2024 Asia-Europe-Pacific School of High-Energy Physics](#)

23 Jul–1 Aug | Knowledge Sharing | European Scientific Institute (FR) | I.FAST | EN
[Accelerators for healthcare? 10-day innovation challenge](#)

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN
[Introduction to Accelerator Physics](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Geneva/Grand-Saconnex: Commissioning of the Route des Nations Monday 25 March

From Monday, March 25, 2024, the new Route des Nations will allow for:

- a new link between the airport, Place des Nations and the heart of International Geneva
- the complete reconfiguration of the Grand-Saconnex motorway junction
- the extension of the tramway from Place des Nations to Grand-Saconnex and on to Ferney-Voltaire.

For more information on the new Route des Nations, click on the following links:

<https://www.ge.ch/document/commissioning-route-nations-major-new-route-getting-around>
[Mise en service de la route des Nations :](#)

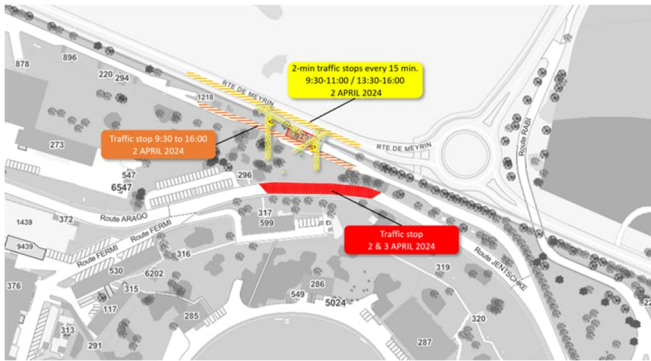
[principes d'accessibilités en voiture | ge.ch](#) (by car/French only)

[Mise en service de la route des Nations :](#)
[principes d'accessibilités en transport public | ge.ch](#) (by public transport/French only)

[Mise en service de la route des Nations :](#)
[principes d'accessibilités en mobilités actives | ge.ch](#) (bicycle and pedestrian paths/French only)

Host State Relations service

Work on the Route de Meyrin: traffic disruptions on 2 and 3 April



SCE department has been informed that a works permission with traffic disruption on Meyrin road has just been granted by the French authorities. From Tuesday 2 April at 9 a.m. to Wednesday 3 April 2024 4 p.m., works on green spaces will take place along the Schrödinger road (inside CERN site) and Meyrin Road (French side).

These works together with a crane installation and operation in the area will cause traffic disruption on adjacent roads, and temporary detour roads.

The Jentschke road will be closed for two days from Tuesday 2 to Wednesday 3 April.

The Schrödinger road will be closed on Tuesday 2 April at 9:00 am until 4:00 pm.

The Meyrin road will remain open to traffic, with temporary two-minute closures every 15 minutes, on Tuesday 2 April from 9:00 am to 4:00 pm.

It is therefore strongly recommended to avoid using these roads and prefer bypasses as far as possible for the whole duration of the works.

Thank you for your understanding.

SCE department

Spring has arrived early, so have the ticks!

A guide to ticks, risks and prevention, to help you enjoy the great outdoors safely

Spring has come early and with warmer days, so have the ticks and the chances of being bitten when out and about. Ticks are little parasitic mites, whose bites can have severe consequences for our health through the transmission of various infectious agents. The most common infections caused by tick bites are [Lyme disease](#) (Lyme Borreliosis), generally treatable with antibiotics, and [tick-borne encephalitis](#) (TBE), which is rarer than Lyme disease with 5000 to 13 000 cases reported globally each year. Although there is no vaccine against Lyme disease, one does exist

against TBE and it is recommended for anyone residing in or travelling to areas where the disease is prevalent. In Europe, the TBE vaccination is recommended in Austria, Czech Republic, Estonia, Finland, Germany, Latvia, Lithuania, Poland, Slovakia, Slovenia, Sweden, Switzerland and Western Russia.

Find out more about ticks, vaccination possibilities, how to protect yourself and what to do if you are bitten by a tick on the dedicated Medical Service webpage: <https://hse.cern/ticks>

Ombud's corner

Difficult conversations (part I): “what happened?”

We all have to face difficult conversations, i.e. anything that we find hard to talk about, whether in our private life or in the workplace. Such conversations are a normal part of life. Often, visitors come to the Ombud because they need to have such a conversation but don't know where to start, how to get their messages across or how to have the conversation without damaging a work relationship.

Here are some examples:

Peter* is made increasingly uncomfortable by the attitude of a colleague, Michael*, who has recently started to criticise him openly and to isolate him from the rest of the team. A conversation is needed but Peter fears his colleague's reaction.

Jenny* needs a serious talk with one of her supervisees, Mona*, who not only fails to deliver according to expectations but also makes disparaging comments about other team members' contributions.

Andrzej* is working hard on his PhD and needs feedback from Tom*, his supervisor. Despite being asked several times, Tom neglects to give proper feedback. Andrzej is at a loss and needs to have a heart-to-heart discussion with Tom.

The book *Difficult Conversations*^[1] – a great read – claims that every difficult conversation that we are worried about having actually unfolds into three intertwined conversations:

- the “what happened?” conversation: a disagreement about what *happened*
- the “feelings” conversation: what are the *feelings and emotions* at stake?
- and the “identity” conversation: what does this say *about me* and about the other party?

Their analysis resonates very much with what I experience in the Ombud Office. In this and the next two articles, I propose to take you through these three conversations and explore with you what is really at stake in each of them.

The “what happened” conversation is the most evident and eats up time in a difficult conversation. In the three example cases that I mentioned, the parties will spend a lot of time disagreeing about what happened, who said what and who did what, who is right, who meant what and who is to blame. Disagreeing is not a bad thing but, in difficult

conversations, the disagreement seems to be at the heart of what is going wrong.

When preparing for their difficult conversations, Peter, Jenny and Andrzej need to remember that each of us sees the world differently and that no one holds the Truth. We have some limited available information, on the basis of which we make observations. Then we interpret those observations, influenced by our past experiences, by our own implicit rules and, very naturally, by our self-interest. No wonder that our stories collide into a difficult conversation, simply because the two stories are different.

The second error that we all make in a budding conflict is to make assumptions about the other party's intentions. Peter assumes that Michael suddenly has this attitude because they are competing for the same position. Jenny assumes that Mona has an inflated view of her abilities. And Andrzej assumes that Tom gives no importance at all to the thesis he is working so hard on. The fact is, we do not know what the other party's intentions are. But we assume what their intentions are, based on the impact that their behaviour has on us (if I feel bad it is because they had bad intentions) and we judge them (they are a bad person), based on the intentions we think we feel certain they have. Conversely, the other party undoubtedly attributes to you intentions that you do not have, based on the impact your actions have had on them, and judges you based on these supposed intentions. Yet another reason for the conversation to go wrong.

The third error we make in the “what happened?” conversation is to focus on who is to blame. Who is the bad person? Who made the mistake? Who should apologise and who gets to be righteously indignant? The problem with focusing on blaming is that it triggers defensive reactions from the other party, gets in the way of the communication and prevents us from understanding what the problem really is, how we may have contributed to it and how to correct it.

With so many pitfalls in a difficult conversation, how can we have a conversation that allows both parties to move forward?

The authors of *Difficult Conversations* propose an interesting solution: they propose to place a third story at the centre of the conversation. Not your story, nor the other party's story, but a story that both parties are ready to discover together and accept:

The first step to the third story is to let go of certainty and try curiosity. If Peter, Jenny and

Andrzej start the conversation by being genuinely curious about how the other party sees things, and why they may see things this way, they have a head start for an effective and open conversation.

The second step to get to the third story is to stop assuming what the other party's intentions are. Remind the person of what they did and what impact this has had on you; point out that you do not know what their intentions were. Very importantly, reflect on the complexity of your own intentions and be honest about them.

The third step to get to the third story is to stop blaming – which gets you nowhere – and to reflect together on your respective contribution(s) to the problem. Reflect with an open mind on how you may be part of the problem.

Easier said than done, but, as CERN Ombud, I find this makes a lot of sense as I observe these mechanisms every day and they become particularly evident during mediation. I would like to encourage you to try

these principles when you engage in what you fear will be a difficult conversation. You might not get to the third story the first time, but the conversation will take you and the other party much closer to a solution. The Ombud can help prepare for a difficult conversation.

In the next article, I propose to explore with you how feelings and emotions play an important part in the conversations that we find so difficult to have and how to untangle that part of the conversation.

Laure Esteveny

** Names and situations are fictitious.*

*This article is inspired by the book *Difficult Conversations* by Douglas Stone, Bruce Patton and Sheila Heen, which I strongly recommend. It is available to loan from the CERN Library.*

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at <https://mattermost.web.cern.ch/cern-ombud/>.

^[1] *Difficult Conversations* by Douglas Stone, Bruce Patton and Sheila Heen